

OmniChannel AI Sales Agent

Intelligent WhatsApp Automation for Lead Qualification & CRM Integration

End-to-End Automation • Multi-Format Message Handling • AI-Powered Conversations

Project Overview

This project delivers a fully automated, AI-powered sales agent that operates through WhatsApp Business API. The system handles inbound customer inquiries across multiple message formats (text, voice, image, video, and document), qualifies leads through intelligent conversation, and syncs qualified prospects directly into Salesforce CRM.

Built using n8n workflow automation, Google Gemini AI models, Supabase vector search, and the WhatsApp Cloud API, the solution replaces traditional manual lead qualification with a 24/7 conversational AI that engages prospects naturally, retrieves accurate product information from a knowledge base, and captures structured lead data without human intervention.

Attribute	Details
Role	AI Automation Architect
Industry	Free Zone / Business Licensing (UAE)
Platform	n8n (Self-Hosted Workflow Automation)
AI Model	Google Gemini 2.5 Pro & Flash
CRM	Salesforce
Channel	WhatsApp Business API (Cloud API)
Knowledge Base	Supabase Vector Store + Cohere Embeddings
Memory	PostgreSQL Chat Memory (per-session)

System Architecture

The workflow follows an event-driven architecture triggered by incoming WhatsApp messages. Each message is classified by type and routed through specialized processing pipelines before reaching the AI agent for intelligent response generation.

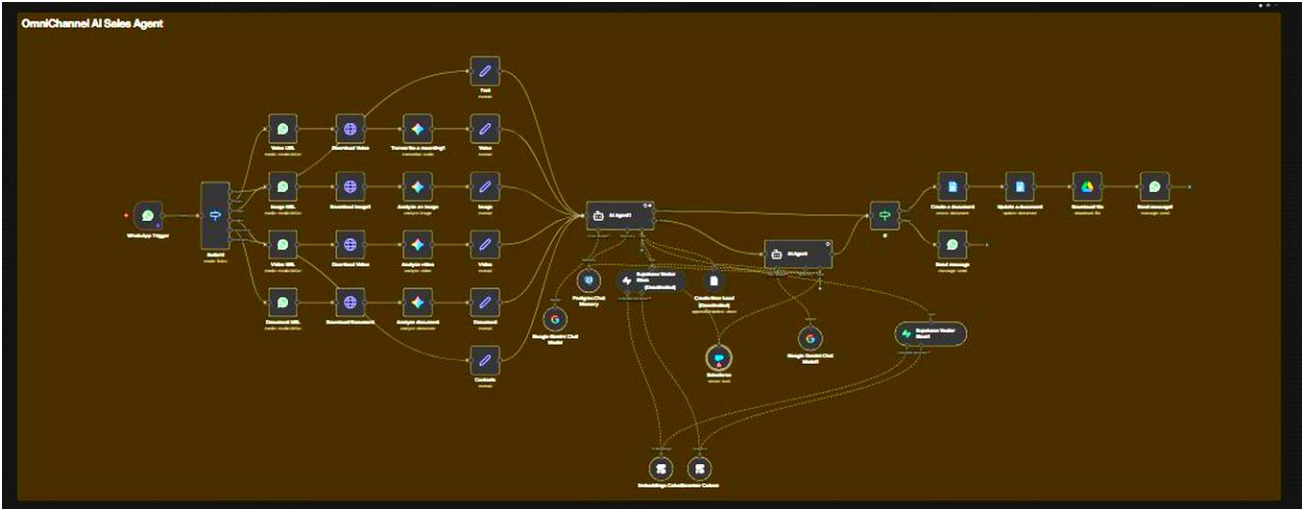


Figure 1: n8n Workflow — OmniChannel AI Sales Agent

High-Level Flow

1. **Inbound Message Reception** — WhatsApp Trigger webhook receives all incoming messages in real time.
2. **Message Type Classification** — A Switch node routes messages by type: Text, Voice, Image, Video, Document, or Contact Card.
3. **Media Processing** — Non-text messages are downloaded via Meta API and processed through Gemini Flash for transcription/analysis.
4. **AI Agent Processing** — The primary AI Agent (Gemini 2.5 Pro) handles conversation with access to vector search and CRM tools.
5. **Fallback Agent** — If the primary agent fails, a secondary agent with identical tools ensures zero message loss.
6. **Response Delivery** — Responses under 4,000 characters are sent as text; longer responses are converted to a document file and sent as an attachment.
7. **Lead Capture** — Qualified prospect data is pushed to Salesforce with proper status tagging.

Multi-Format Message Handling

A core differentiator of this system is its ability to process six distinct message types, converting all inputs into text that the AI agent can reason over.

Message Type	Processing Pipeline	Output
Text	Direct extraction from payload	Raw text body passed to agent
Voice/Audio	Download → Gemini Flash transcription	Transcribed text

Image	Download → Gemini Flash image analysis	Image description + caption
Video	Download → Gemini Flash video analysis	Video description + caption
Document	Download → Gemini Flash document parsing	Document summary + caption
Contact Card	Direct extraction from vCard payload	Name, phone, WhatsApp ID

AI Agent Design

The conversational AI agent is designed as a professional sales consultant. It engages prospects through natural, consultative dialogue while leveraging two primary tools to deliver accurate information and capture leads.

Agent Tools

Supabase Vector Store (RAG): Connected to a Supabase vector database containing comprehensive product documentation. Uses Cohere multilingual embeddings for semantic search and Cohere reranker for precision. The agent queries this tool to verify business activity availability, retrieve accurate pricing, explain policies, and answer FAQs with grounded information.

Salesforce Lead Capture: An append-or-update tool that writes qualified lead data to Salesforce using phone number as the matching key. The agent autonomously decides when to trigger lead capture based on conversation signals such as pricing inquiries, explicit interest statements, or requests for follow-up contact.

Conversation Memory

The system uses PostgreSQL-backed chat memory with a 20-message context window, keyed by the sender's WhatsApp ID. This enables the agent to maintain coherent multi-turn conversations across sessions, recall previously shared information, and avoid redundant questions.

Dual-Agent Fallback

The architecture implements a primary-secondary agent pattern. If the primary Gemini 2.5 Pro agent encounters an error (API timeout, safety filter, malformed response), the workflow automatically routes the message to a fallback agent with identical tools and memory access. This ensures zero message loss and continuous customer engagement.

Lead Qualification Logic

The agent follows a structured qualification framework before capturing leads, ensuring data quality and proper categorization in Salesforce.

Qualification Criteria

- Business activity must be verified as supported via the vector knowledge base
- Visa eligibility must be assessed based on nationality rules and requirements
- Minimum required fields collected: First Name, Last Name, Email, Phone
- Prospect must demonstrate clear intent (pricing inquiry, setup request, contact request)

Lead Status Classification

Status	Condition
Qualified – Activity Supported & Visa Eligible	Business activity is approved AND visa requirements can be met
Unqualified – Activity Not Supported	Requested business activity is not available in the free zone
Unqualified – Visa Restriction	Activity is supported but nationality has visa restrictions for new applications

Importantly, unqualified leads are still captured in Salesforce with the appropriate status tag, enabling the sales team to follow up with alternative solutions or track demand for unsupported activities.

Smart Response Delivery

The system includes an intelligent response length handler. If the AI agent generates a response exceeding 4,000 characters (WhatsApp's practical display limit), the workflow automatically creates a Google Doc, writes the full response, downloads it as a file, and sends it as a WhatsApp document attachment with a summary caption. Shorter responses are sent as standard text messages.

Technology Stack

Component	Technology
Workflow Engine	n8n (self-hosted)
AI Models	Google Gemini 2.5 Pro (conversation) + Gemini 2.5 Flash (media processing)
Messaging API	WhatsApp Business Cloud API (Meta)
Vector Database	Supabase (pgvector)
Embeddings	Cohere embed-multilingual-v3.0

Reranker	Cohere rerank-multilingual-v3.0
Chat Memory	PostgreSQL (n8n Postgres Chat Memory node)
CRM	Salesforce (lead capture & management)
Document Generation	Google Docs API + Google Drive API
Authentication	OAuth 2.0 (Google), API Keys (Meta, Cohere, Gemini)

Lead Data Fields Captured

The following fields are captured in Salesforce for each qualified lead:

Field	Source	Required
First Name	AI-extracted from conversation	Yes
Last Name	AI-extracted from conversation	Yes
Email	AI-extracted from conversation	Yes
Phone / WA Number	Auto-captured from WhatsApp payload	Yes (auto)
Company Name	AI-extracted	No
Business Activities	AI-extracted	No
Entity Type	Individual / Corporate	No
Package Interest	AI-recommended or user-selected	No
License Period	1–10 years	No
Office Space Requirements	Shared / Dedicated / Custom	No
Visa Requirements	Number needed or None	No
Category	B2B / B2C / B2G	No
Lead Status	AI-determined qualification status	Yes

Key Features & Highlights

- 24/7 autonomous lead qualification without human intervention
- Six message types supported: text, voice, image, video, document, and contact cards
- RAG-powered responses grounded in real product data via Supabase vector search
- Multilingual support through Cohere multilingual embeddings and reranker
- Per-session conversation memory with 20-message context window

- Dual-agent failover architecture ensuring zero message loss
- Smart response delivery: text for short replies, document attachments for long ones
- Structured lead capture with qualification logic and status tagging in Salesforce
- Content safety filters applied across all AI model interactions
- Scalable architecture: handles concurrent conversations across unlimited users

Results & Impact

- Eliminated manual lead screening, reducing response time from hours to seconds
- Enabled 24/7 prospect engagement across all time zones
- Automated multi-format message processing removed the need for separate voice/image handling workflows
- Structured Salesforce integration provided the sales team with pre-qualified, enriched lead data
- Fallback agent architecture achieved near-100% uptime for customer-facing interactions

Built by Khaled Salah

AI Automation Developer